05 · Risk register — the honest wall

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> *Confidential DD dossier. The substrate's own tradition: state the limits as loudly as the vision, or it's marketing. This is the section diligence should weight most. Nothing here is hidden elsewhere; it is collected for scrutiny.*

Technical risk

R1 — The energy model is a router, not a mind (the core bet). Today's model is a cosine-to-centroid plus a thin linear head — soft priors over "what resolved before." Real reasoning is **not** in this model. The leap from "thin router" to "intelligence" is *unbuilt* and comes from one of two unproven places: richer composed energy models (which return some of the compute cost) or emergent collective composition (many thin routers + verification composing into reasoning). **This is the whole company's central technical bet.** Mitigation: the falsifiable-atom discipline — prove each rung (the router *is* proven) before funding the next. Do not mistake the scaffold for the building.

R2 — Retrieval ≠ reasoning. Indexing the world gives knowledge *access*, not thought. Superintelligence needs composition, planning, and verification *over* retrieved knowledge. The energy-descent-as-generation primitive is the seed, but it is linear and shallow. **The reasoning layer is the hard, unsolved core.**

R3 — Verification is the crux and the single biggest risk. A system that learns on-fire from verdicts is only as good as its verdicts. Bad or *gamed* verdicts don't just add noise — they train the shared prior toward delusion. The guard (a canonical, fixed reference anchor + independent multi-party corroboration + the reward economics' verification gate) is **load-bearing**: if it fails, the collective becomes confidently, collectively wrong. This is the line between wisdom-of-crowds and madness-of-crowds, and it is **not automatic.**

R4 — "Low compute" is low *marginal*, not low *total*. Embedding and serving retrieval over the live web is heavy infrastructure. The compute *moves* from training to indexing/serving; it does not vanish. Honest claim: cheap-per-thought, not cheap-overall. The "half the cost" pitch is a per-task target, not a total- infrastructure claim, and is **not yet independently benchmarked.**

R5 — Energy containment (the founder's own stated blocker). The design is power-hungry — it "leaks energy" by nature (it trains and grows storage on every fire). This is *why* it isn't released yet. The on-chain offload is the intended fix (weights on-chain, laptop as frecency cache), and is DESIGNED but not proven at scale. Until contained, the product is unsafe/uneconomical to hand to users.

R6 — Query privacy is unsolved; ZK is unbuilt. Data-at-rest is sealed, but querying a shared index can leak intent (needs PIR/local-replicas/ZK). And the headline "ZK credential binding" is a **direction, not a shipped feature** — today's identity layer is hash commitments + signed lineage. Marketing that says "privacy you can verify" is ahead of the code; diligence should hold the code's actual maturity (see `03-ONCHAIN-privacy-solana.md`).

Strategic / market risk

R7 — The algorithm is copyable; the moat is the network, not the code. If the EBM harness works, frontier labs can and will build it — it is proprietary but not un-reproducible. The defensibility thesis is **distribution + the ledger** (Solana's storage + wallet distribution + a commons of corroborated weights won first). That is a race, not a lock. If Aiko does not win the network before the approach is commoditized, the moat does not hold.

R8 — Single-user, not yet a collective. The thesis's power (compounding intelligence, shared prior) requires *many* agents firing and corroborating. Today there is **one user** (the founder). Every "collective" property is architectural potential, not observed behavior. The network effects are unproven because the network does not yet exist.

R9 — Pre-launch, zero external validation. Not publicly released, no external users, no revenue. Capability is founder-demonstrated (Overwatch one-shot, ARC-AGI) and awaits independent, recorded replication.

R10 — A new-currency ambition adds scope and regulatory surface. The founder's end-vision includes defining a *new currency that rewards novelty/value-created* (beyond value-of-work). This is early and unformed, and paying token holders from usage introduces securities-like questions. It should be treated as a research

aspiration, not a near-term deliverable, and de-scoped from the core investment thesis.

Execution / key-person risk

R11 — Single founder, deep in a rabbit hole. One person, ~2–3 months into a very deep build, self-aware about scope creep ("dug myself into a deep rabbit hole"). Enormous surface area (harness + model + on-chain + economics) for one person. Key-person risk is maximal; the first hire and a scoped roadmap are the obvious de-risking moves.

R12 — Two projects in superposition. Aiko is a partial pivot from Unbrowse; Unbrowse still exists. Focus risk: the pivot is not clean, and "which company is this" is a real question a first-round panel already flagged.

"Superintelligence" — a bet, not a result

Collective systems (markets, evolution, science) *do* exceed any participant, so the ambition is not crazy. But whether this reaches *super* — and not merely *useful* — is untested and hinges entirely on R1, R2, and R3 holding. It should be underwritten as a **falsifiable direction with a cheap first gate already passed** (the router), not as a destination being claimed.

The one honest sentence

The router is proven; the reasoning is a bet; the network doesn't exist yet; the privacy pillar is real for data and aspirational for ZK; and the moat is a race to own a commons before the idea is copied. Everything compelling about Aiko is downstream of bets that are named, cheap to test next, and being tested in order.